FACULDADE DE CIÊNCIAS E TECNOLOGIA
DEPARTAMENTO DE FÍSICA

Machine Leaning (ML)

BASIC CONCEPTS AND EVALUATION METRICS

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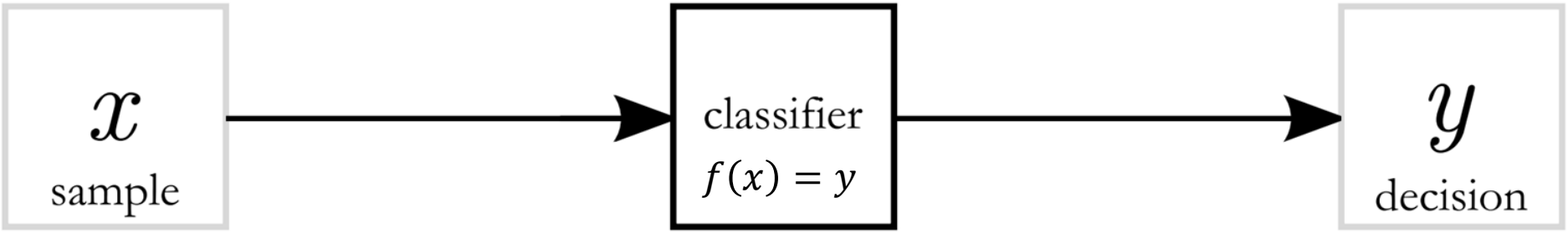
Basic Concepts



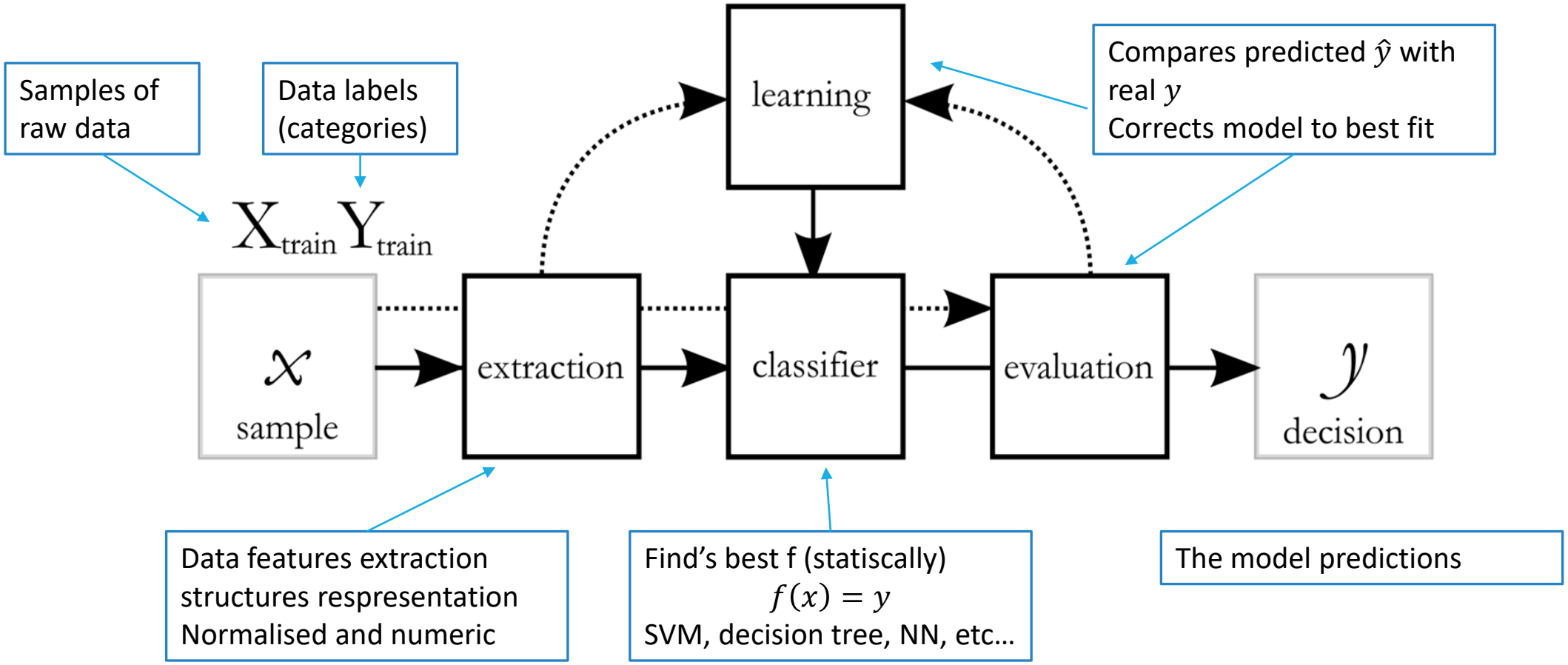
- Input data point:
- Could be as ex: Image, signal, text, etc...
- Feature vector is built
- Labels are generated

- Recognise patterns
- Could be anything from (like logistic regression) or highly complex (like a deep neural network)

- Output labelled class:
- Ex: - “cat”, “dog”, or “bird”

Goal: A classifier that learns the statistical patterns in the training data and is able to generalize its conclusions to new, unseen data.

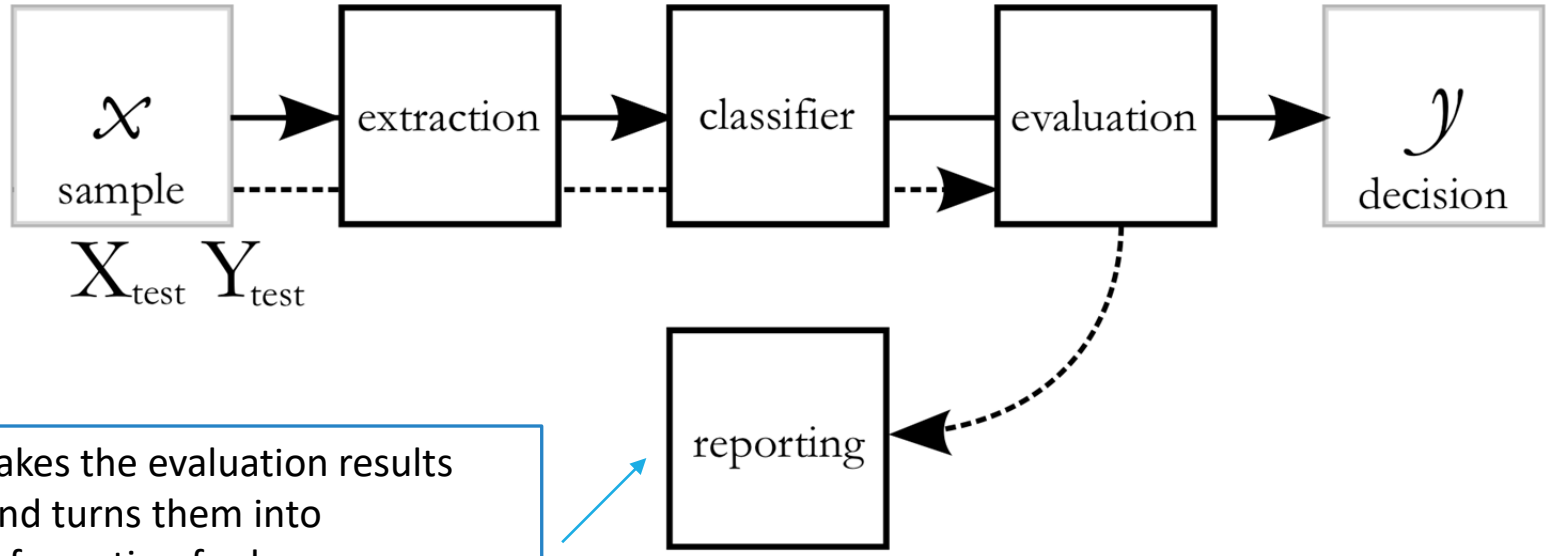
Basic Concepts
Supervised ML Classifier



Basic Concepts
Train and test datasets

But a single data set x_{train}, y_{train} is not enough to guaranty a good model. It is necessary a test data set

Basic Concepts
Evaluation & Reporting



Takes the evaluation results and turns them into information for humans. Accuracy, confusion matrix, …

Basic Concepts
Learning Curves

How do we know that are model is converging for a solution?

The ideal was, at the end, for all the data of our training set: $\mathrm{f}(y_i) = \hat{y}_i = y_i$

So, we can quantify the error with:

$$\mathrm{MSE} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

(Mean Square Error)

$$\mathrm{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|$$

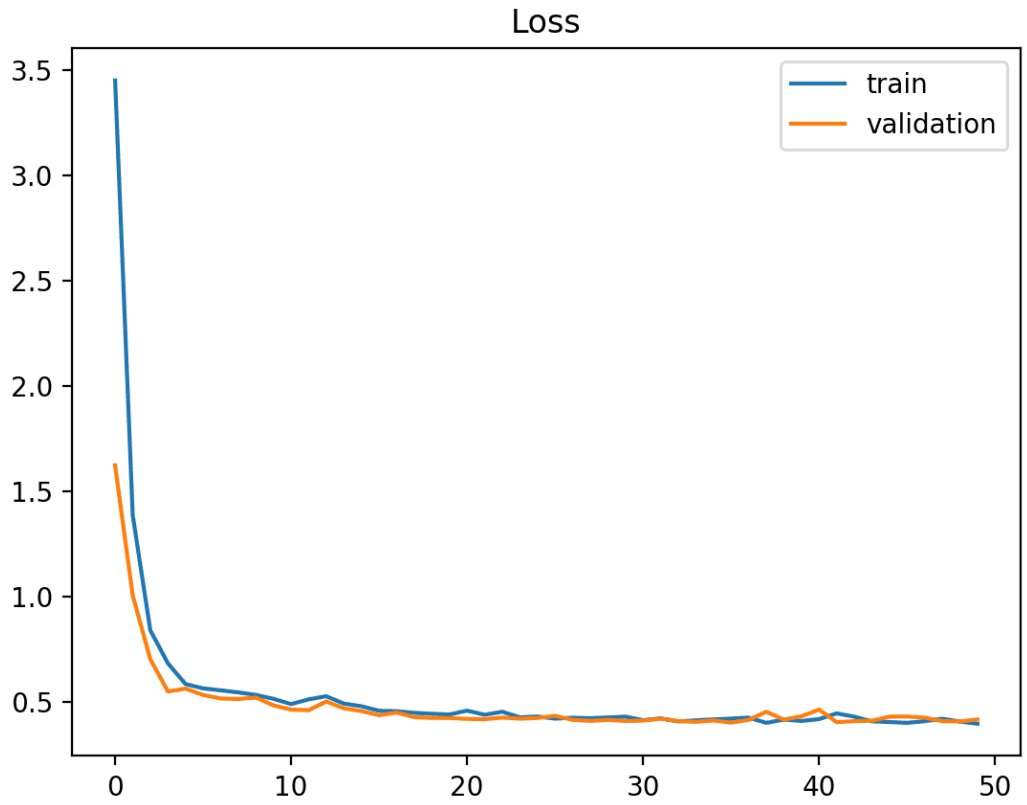
(Mean Absolute Error)

These are know as the loss functions and from them we can built the learning curves.

The efciency of the learning process can be analysed using the training and validation learning curves

The learning curves is one of the simplest and most powerful ways to prove that a supervised ML system is actually learning, but they are not applied to all methods.

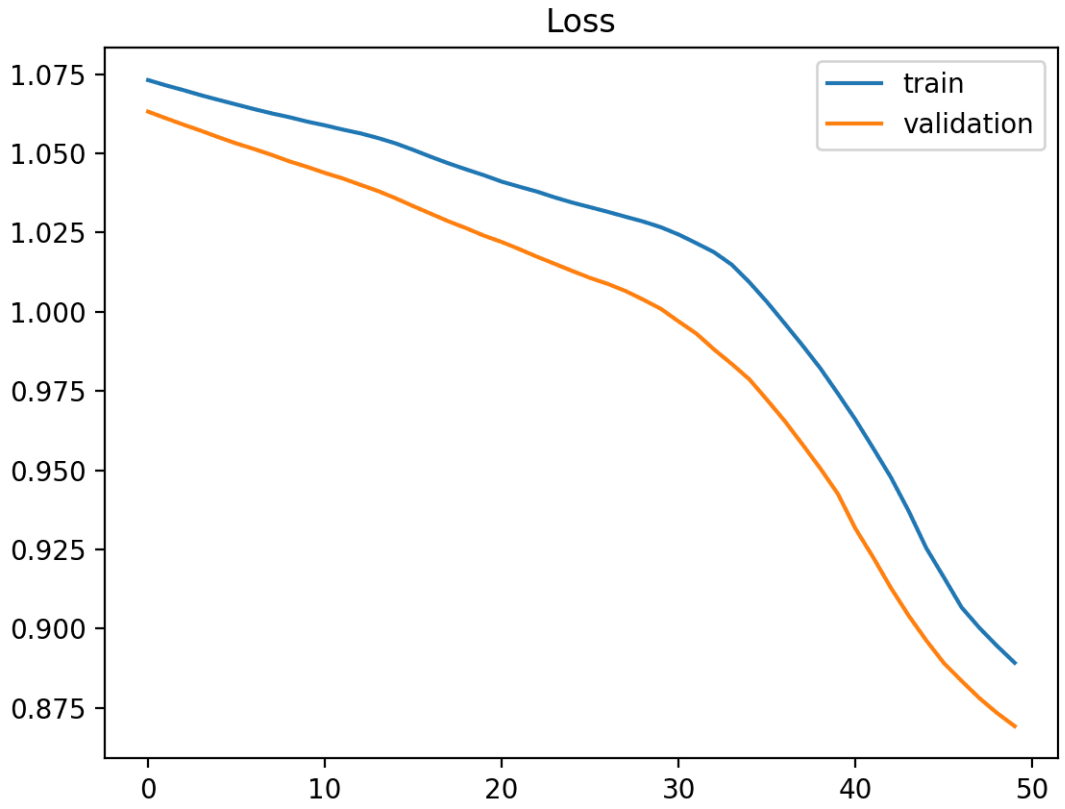
Basic Concepts
Learning Curves – Model Behavior



Good Fit Learning Curves

Train and Validation Learning Curves Showing a Good Fit

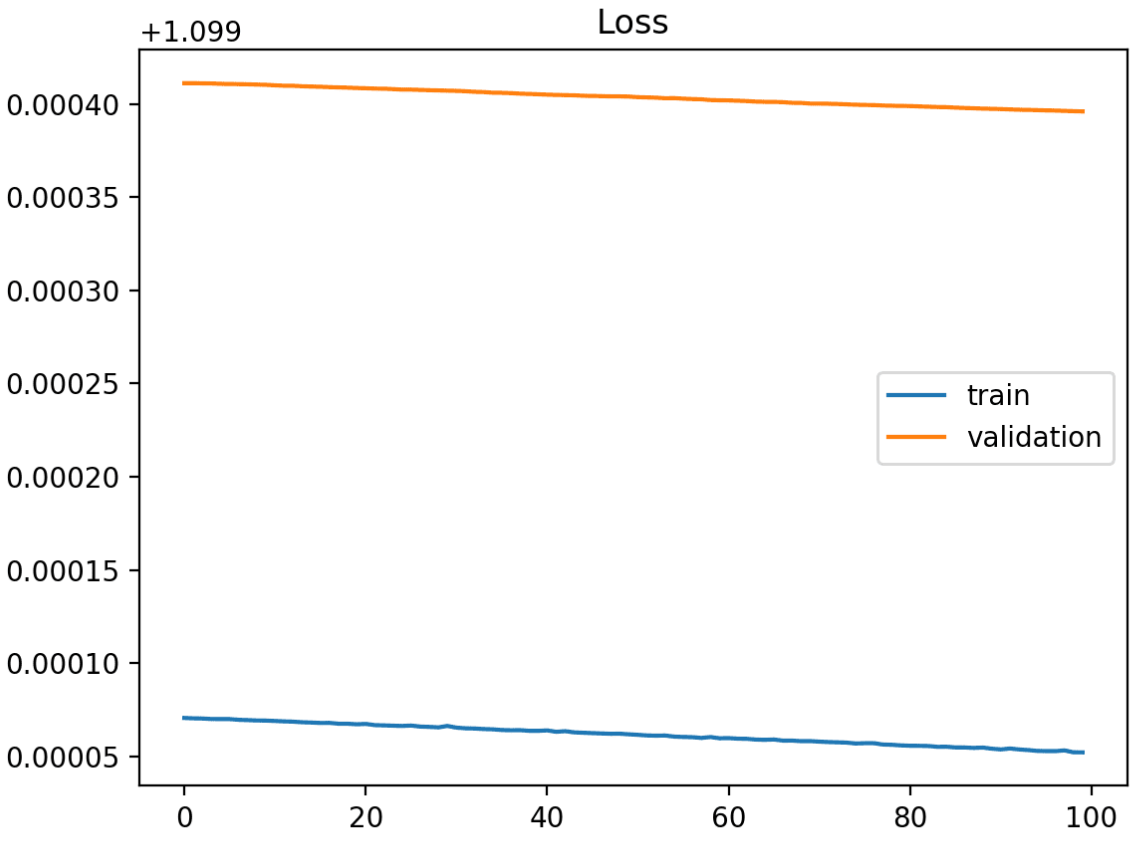
Basic Concepts
Learning Curves – Model Behavior



Underfit Learning Curves

Training Learning Curve Showing an Underfit Model That Requires Further Training

Basic Concepts
Learning Curves – Model Behavior

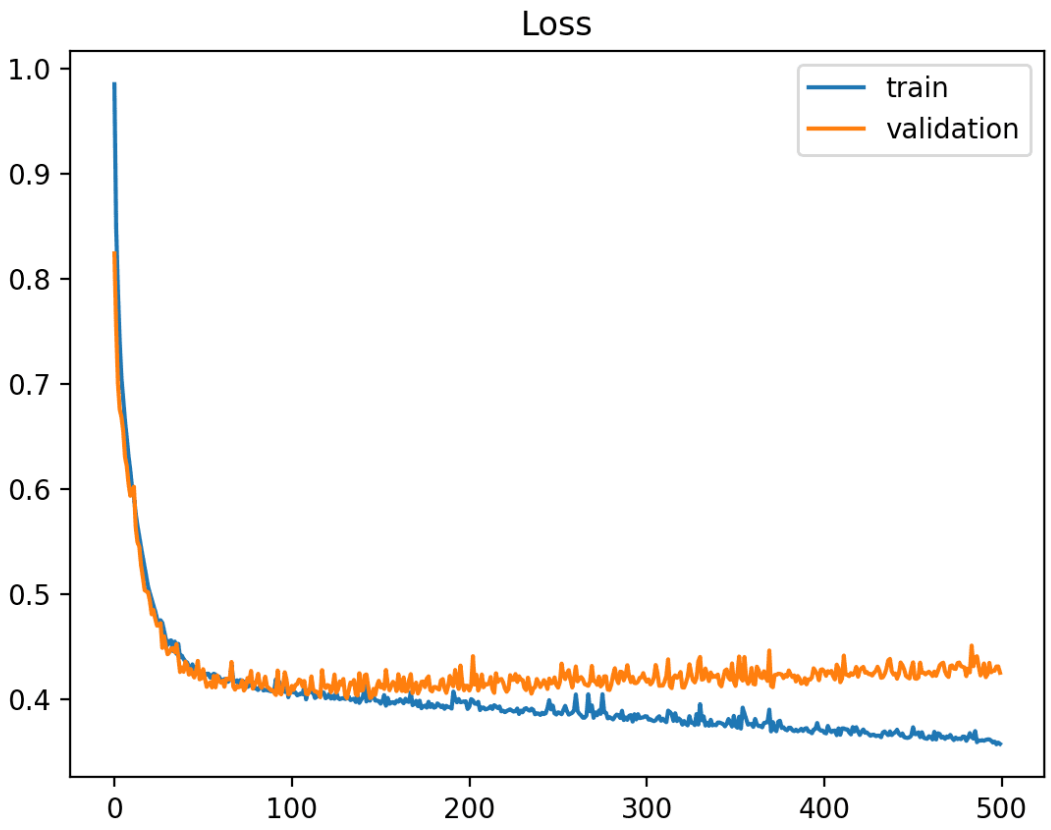


Underfit Learning Curves

Training Learning Curve Showing An Underfit Model That Does Not Have Sufficient Capacity

Underfitting occurs when the model is not able to obtain a sufficiently low error value on the training set.

Basic Concepts
Learning Curves – Model Behavior

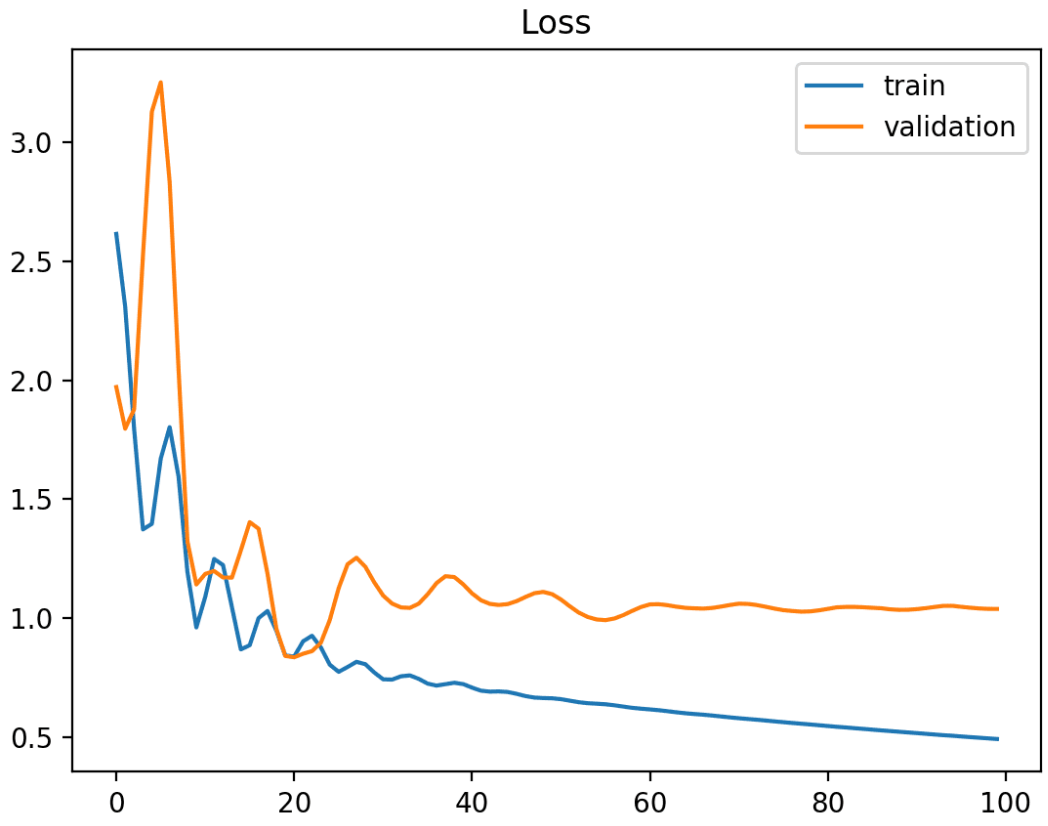


Overfit Learning Curves

Train and Validation Learning Curves Showing an Overfit Model

Overfitting refers to a model that has learned the training dataset too well, including the statistical noise or random fluctuations in the training dataset.

Basic Concepts
Learning Curves – Model Behavior

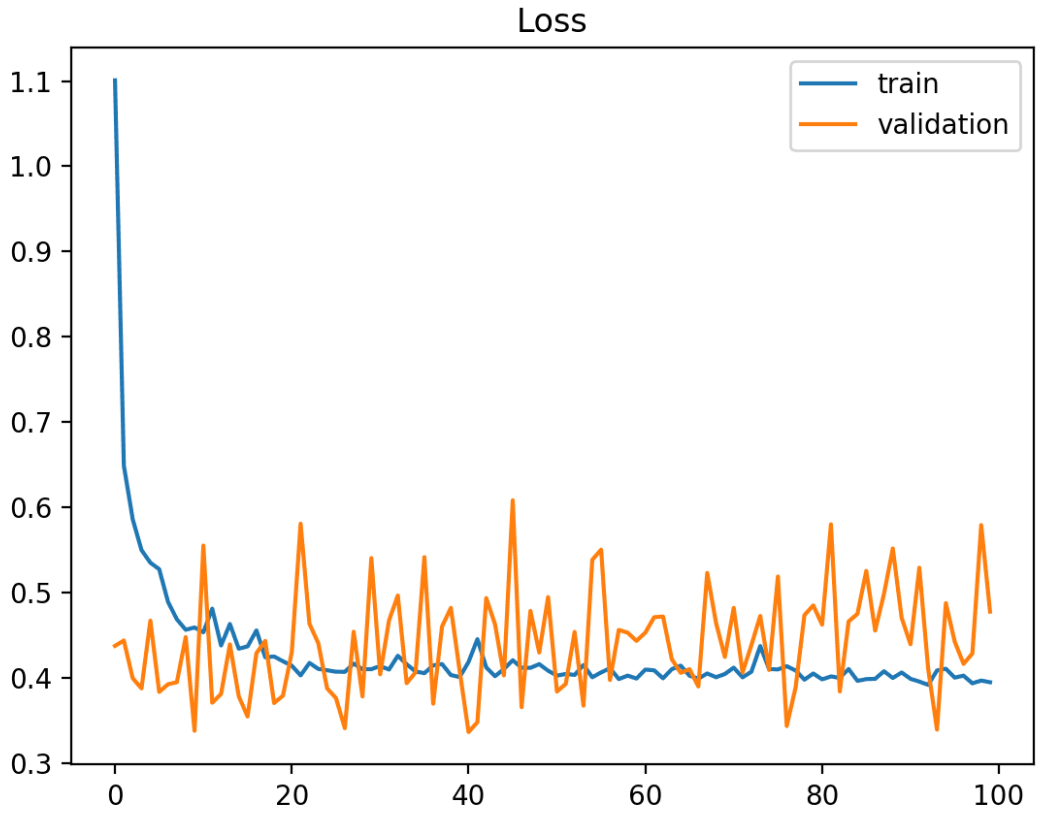


Unrepresentative Train Dataset

Train and Validation Learning Curves Showing a Training Dataset That May Be too Small Relative to the Validation Dataset

An unrepresentative training dataset means that the training dataset does not provide sufficient information to learn the problem, relative to the validation dataset used to evaluate it.

Basic Concepts
Learning Curves – Model Behavior

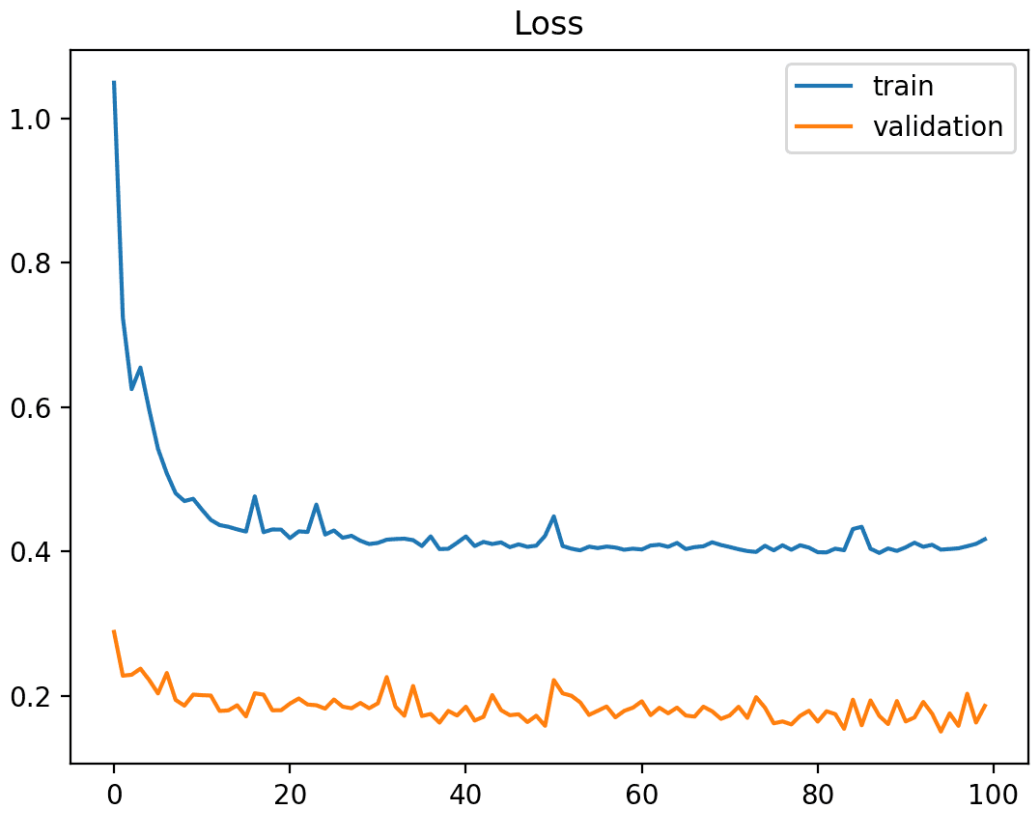


Unrepresentative Validation Dataset

Train and Validation Learning Curves Showing a Validation Dataset That May Be too Small Relative to the Training Dataset

An unrepresentative validation dataset means that the validation dataset does not provide sufficient information to evaluate the ability of the model to generalize.

Basic Concepts
Learning Curves – Model Behavior



Unrepresentative Validation Dataset

Train and Validation Learning Curves Showing a Validation Dataset That Is Easier to Predict Than the Training Dataset

Basic Concepts
Confusion Matrix

		Predicted condition	
	Total population = P + N	Positive (PP)	Negative (PN)
Actual condition	Positive (P)	True positive (TP), hit	False negative (FN), type II error, miss, underestimation
	Negative (N)	False positive (FP), type I error, false alarm, overestimation	True negative (TN), correct rejection

A confusion matrix is the simplest and most powerful ways to understand how a classifier is behaving.

Each cell means:

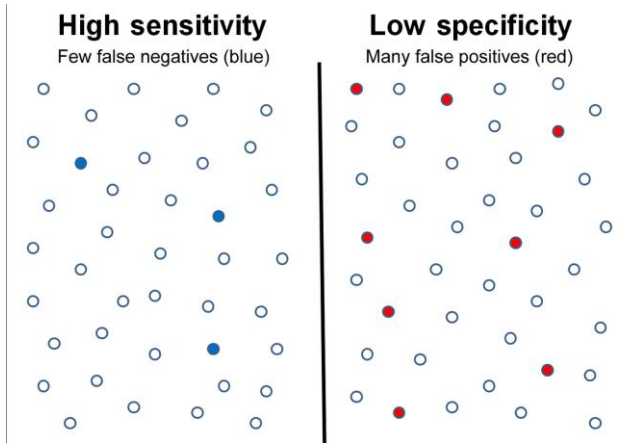
- **True Positive (TP):** model correctly predicted positive
- **True Negative (TN):** model correctly predicted negative
- **False Positive (FP):** model predicted positive but it was wrong
- **False Negative (FN):** model predicted negative but it was wrong

Basic Concepts
Specificity and Sensitivity

Sensitivity, Recall or True Positive Rate

- Of all the real positives, how many did the model catch?
  - Measures how good the model is at finding the positive cases
  - High sensitivity = few false negatives

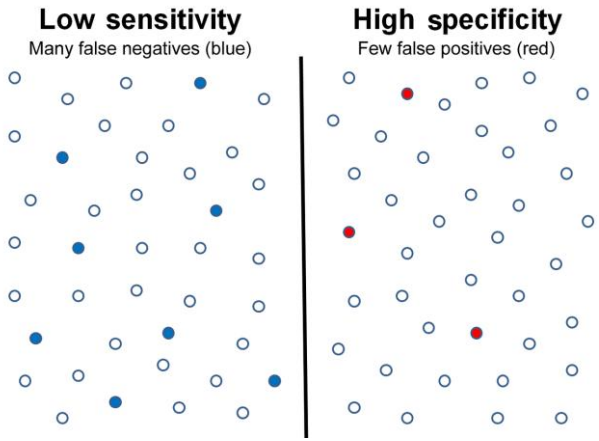
$$\text{Sensitivity} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}}$$



Specificity ou True Negative Rate

- Of all the real negatives, how many did the model correctly reject?
  - Measures how good the model is at avoiding false positives
  - High specificity = few false positives

$$\text{Specificity} = \frac{\text{True Negatives}}{\text{True Negatives} + \text{False Positives}}$$

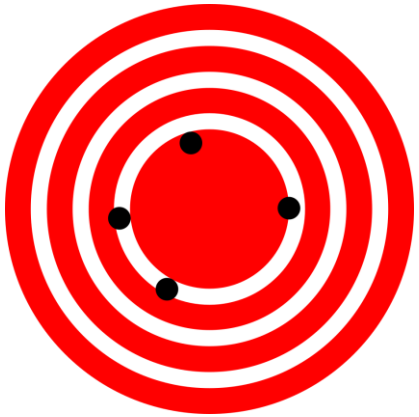


Basic Concepts
Precision and Accuracy

Precision

- When the model predicts positive, how often is it right?.
  - Measures the quality of positive predictions

$$\boldsymbol{Precision} = \frac{True\ Positives}{True\ Positives + False\ Positives}$$



Low accuracy due to low precision

Accuracy

- how often the model is correct overall
  - Just works well only when classes are balanced

$$\boldsymbol{Accuracy} = \frac{True\ Positives + True\ Negatives}{Total}$$

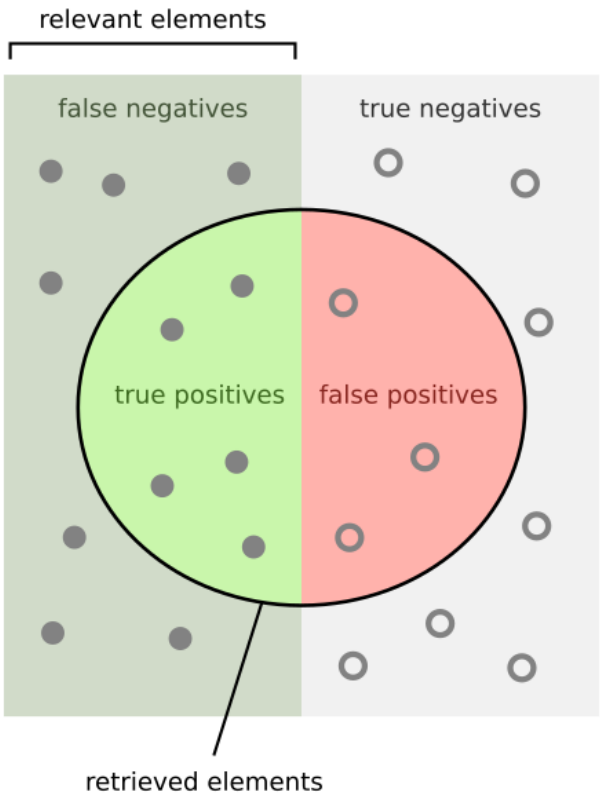
Low accuracy even with high precision

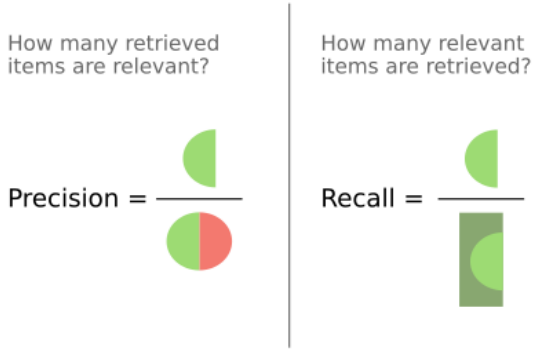
Basic Concepts
F1 Score

F1 Score

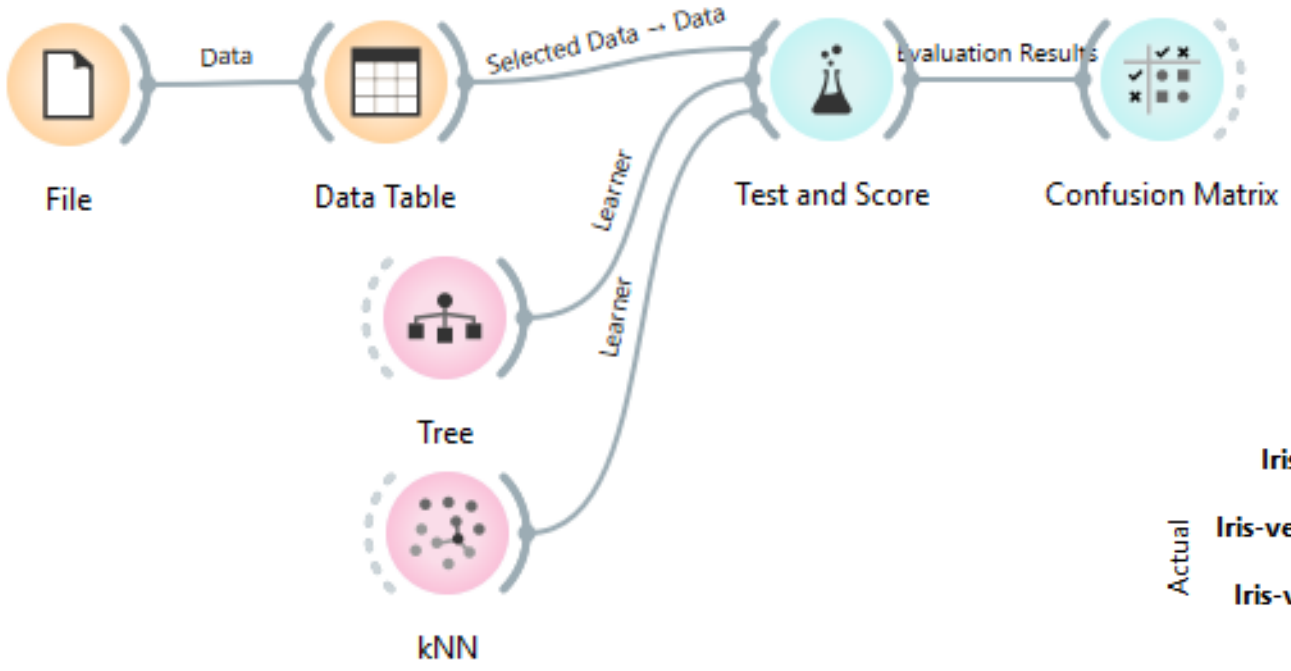
- All previous metrics do o not work well in unbalanced populations
- A single number that balances precision and sensitivity/recall
- Works well even with unbalanced datasets
- Answers: How good is the model when we care about both catching positives *and* avoiding falses

$$F_1 = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$





Basic Concepts
Orange Example



Model	AUC	CA	F1	Prec	Recall	MCC
Tree	0.955	0.927	0.928	0.928	0.927	0.891
kNN	0.990	0.967	0.967	0.967	0.967	0.950

		Predicted			
		Iris-setosa	Iris-versicolor	Iris-virginica	Σ
Actual	Iris-setosa	166	4	0	170
	Iris-versicolor	0	154	16	170
	Iris-virginica	0	17	153	170
	Σ	166	175	169	510

Example_1.ows

Basic Concepts
ROC Curve (Receiver Operating Characteristic)

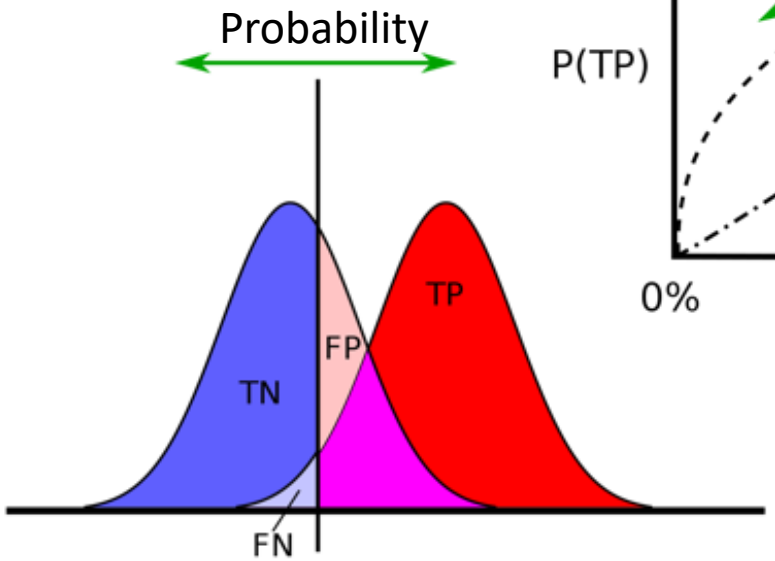
Lets assume a binary classification (ex: survived / not survived)

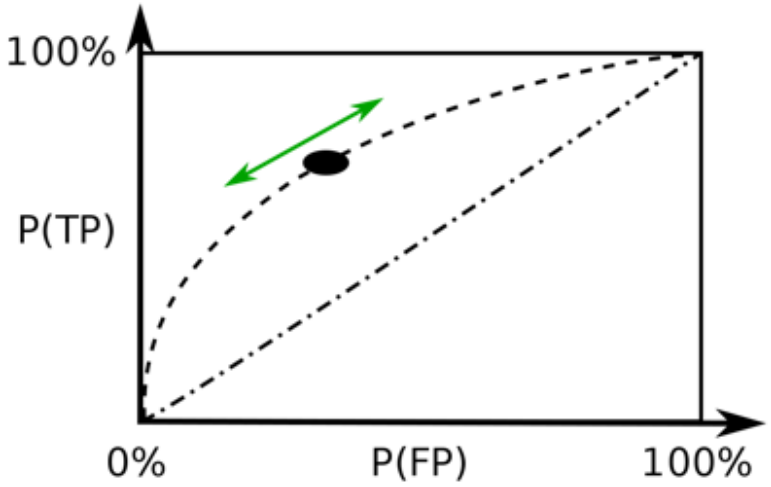
During classification is considered to belong to one of the classes if probabily above some threshold

By default that probability is set to 0.5.

ROC Curve allows to perdict your ratio of true/false positives, depending of the probability

Different classifiers will produce distinct ROC curves





TP	FP
FN	TN

Basic Concepts
ROC Curve (Receiver Operating Characteristic)

Three diferente classfiers

True	Predicted	
	pos	neg
pos	40	60
neg	30	70

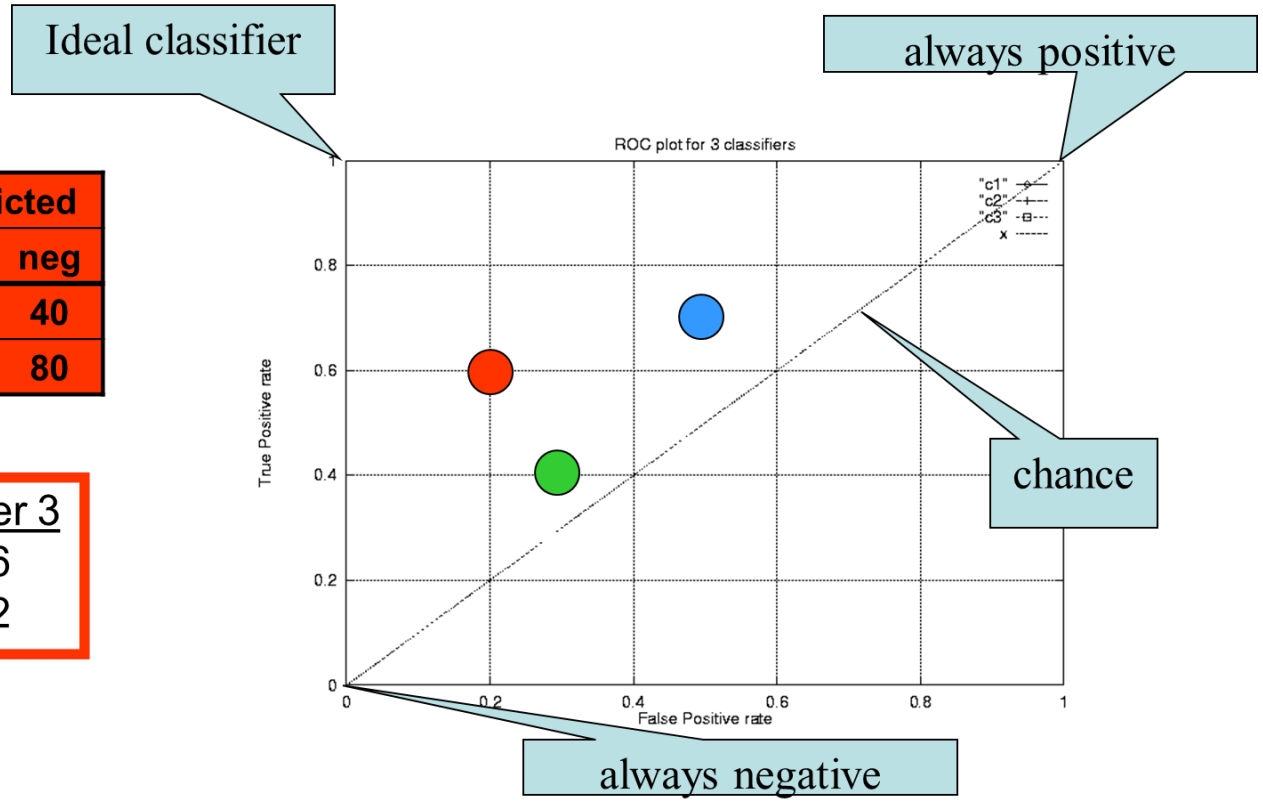
True	Predicted	
	pos	neg
pos	70	30
neg	50	50

True	Predicted	
	pos	neg
pos	60	40
neg	20	80

Classifier 1
TP = 0.4
FP = 0.3

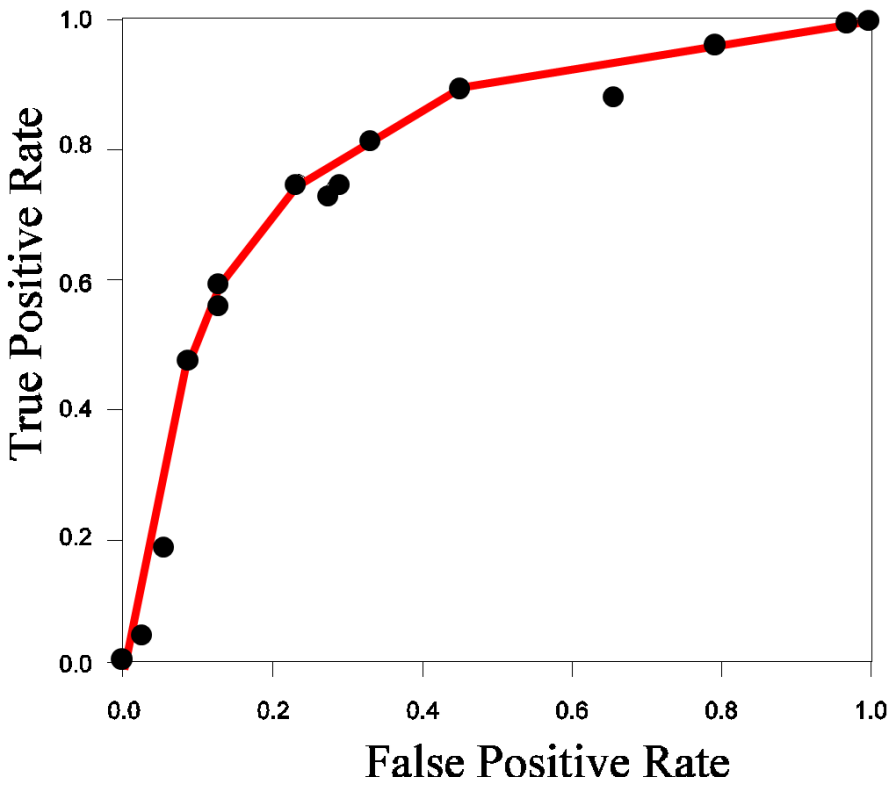
Classifier 2
TP = 0.7
FP = 0.5

Classifier 3
TP = 0.6
FP = 0.2



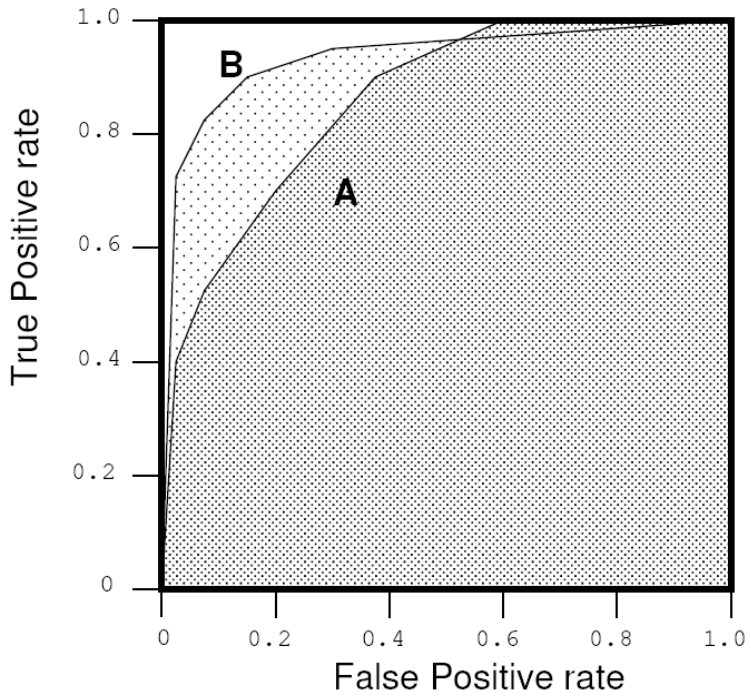
Basic Concepts
ROC Curve (Receiver Operating Characteristic)

A ROC curve is built by analysing the output of the classifier for a wide range of probabities,

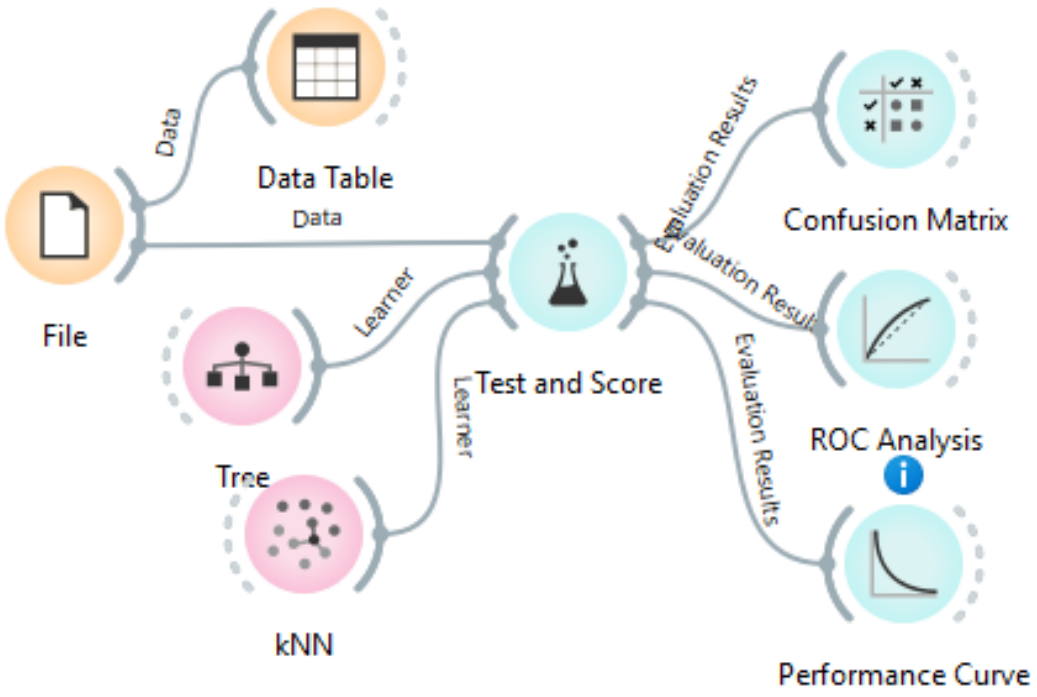
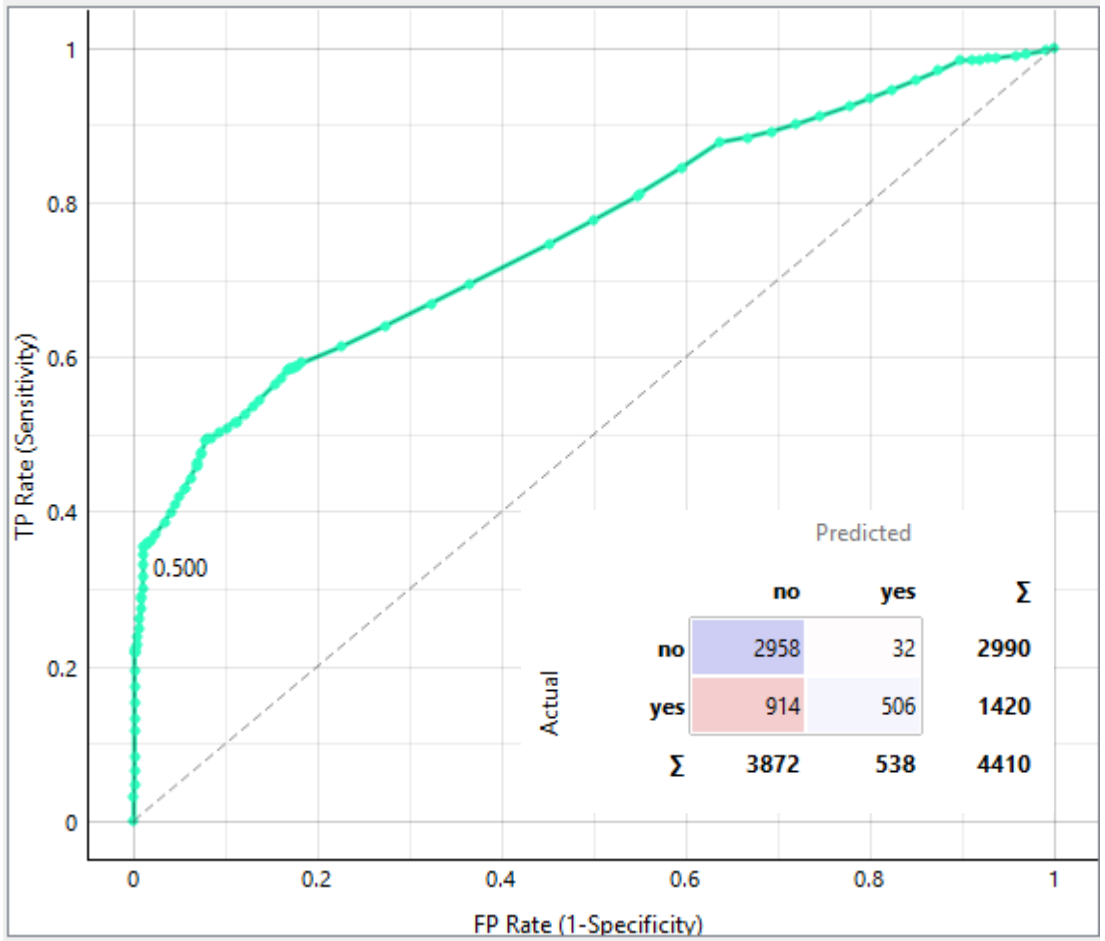


The Area Under the Curve is a good metrics of the performace of your classifier

- 1 perfect fit; 0.5 random fit



Basic Concepts
ROC Curve (Receiver Operating Characteristic)



Example_2.ows